Skill - 1

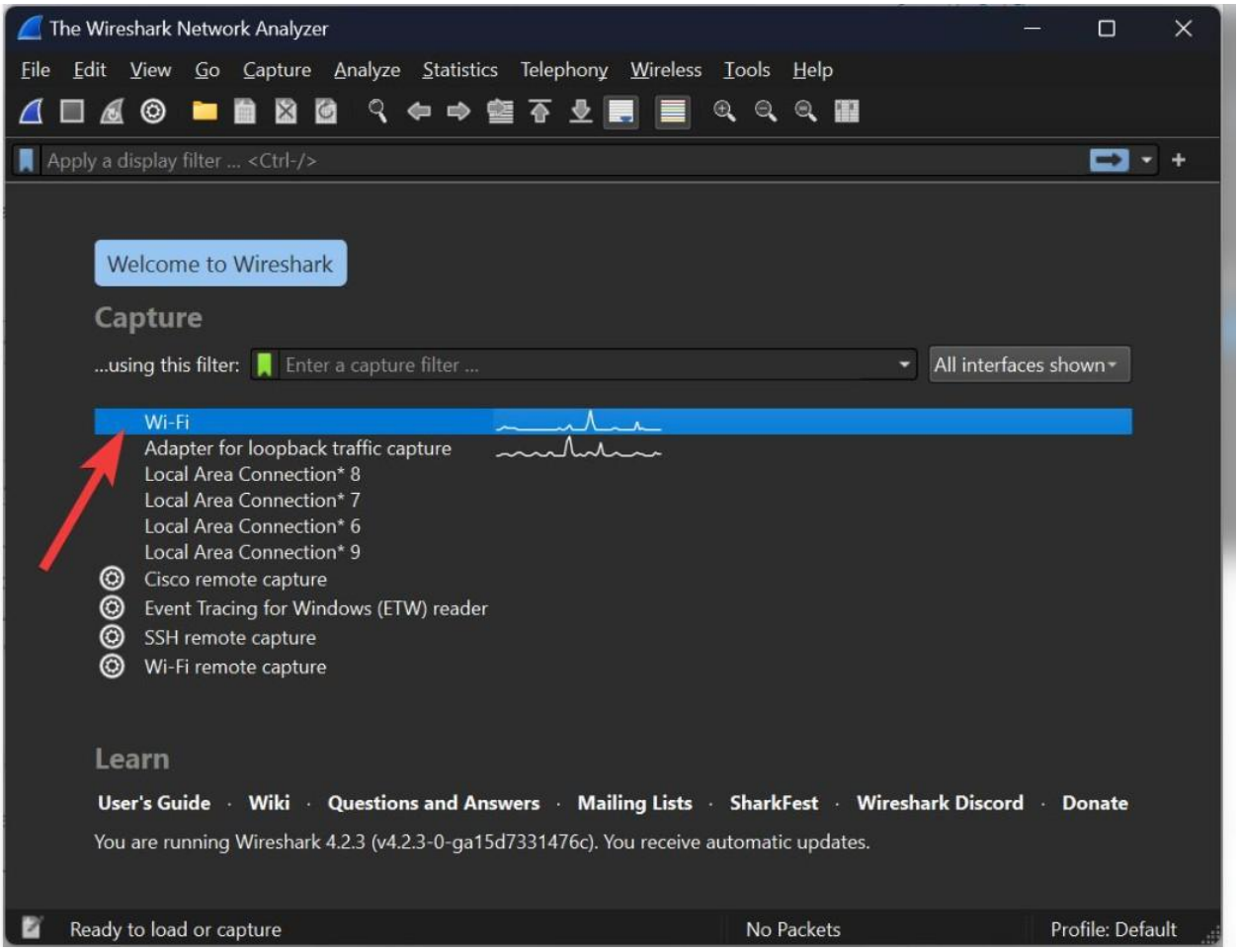
Encapsulation Builder: Analyze the overhead introduced by protocol headers and determine how encapsulation affects network efficiency.

Aim: To capture network packets and understand how Ethernet, IP, and TCP/UDP headers are added during data transmission.

Whenever data travels through a computer network, each protocol layer adds its own header to the data. These headers introduce overhead. Using Wireshark, we will capture real network packets, examine Ethernet, IP and TCP/UDP headers, calculate the total overhead, and determine how efficiently the network carries application data.

Step 1: Install Wireshark

1. Open the browser.
2. Install Wireshark [Wireshark • Go Deep | Download]
3. Install it using the default settings.
4. Open Wireshark.

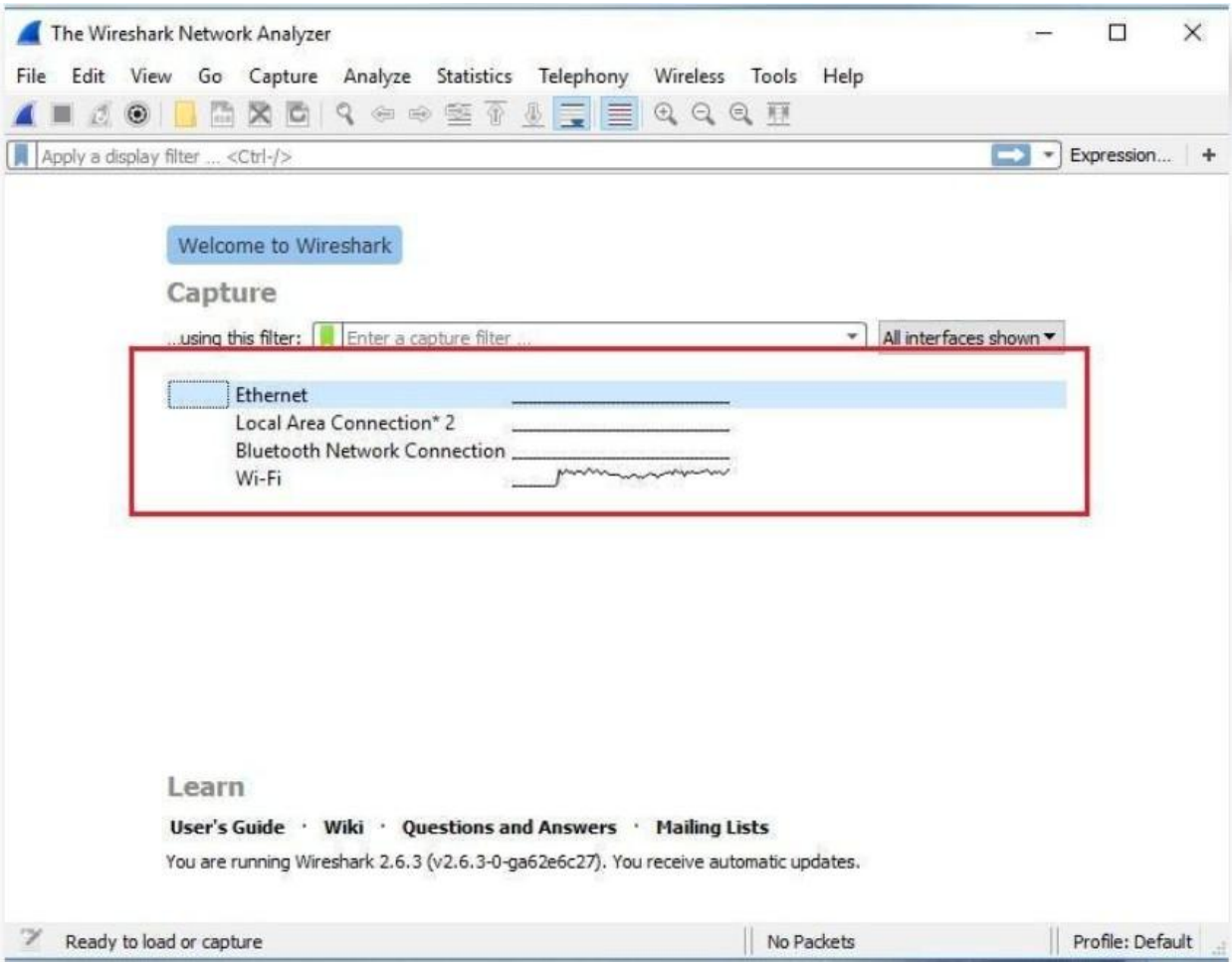


Step 2: Select your network interface

When Wireshark opens, you will see interfaces such as:

Wi-Fi, Ethernet, Bluetooth

1. If you are connected through Wi-Fi, select Wi-Fi.
2. Double-click the Wi-Fi interface.
3. Packet capture will start.

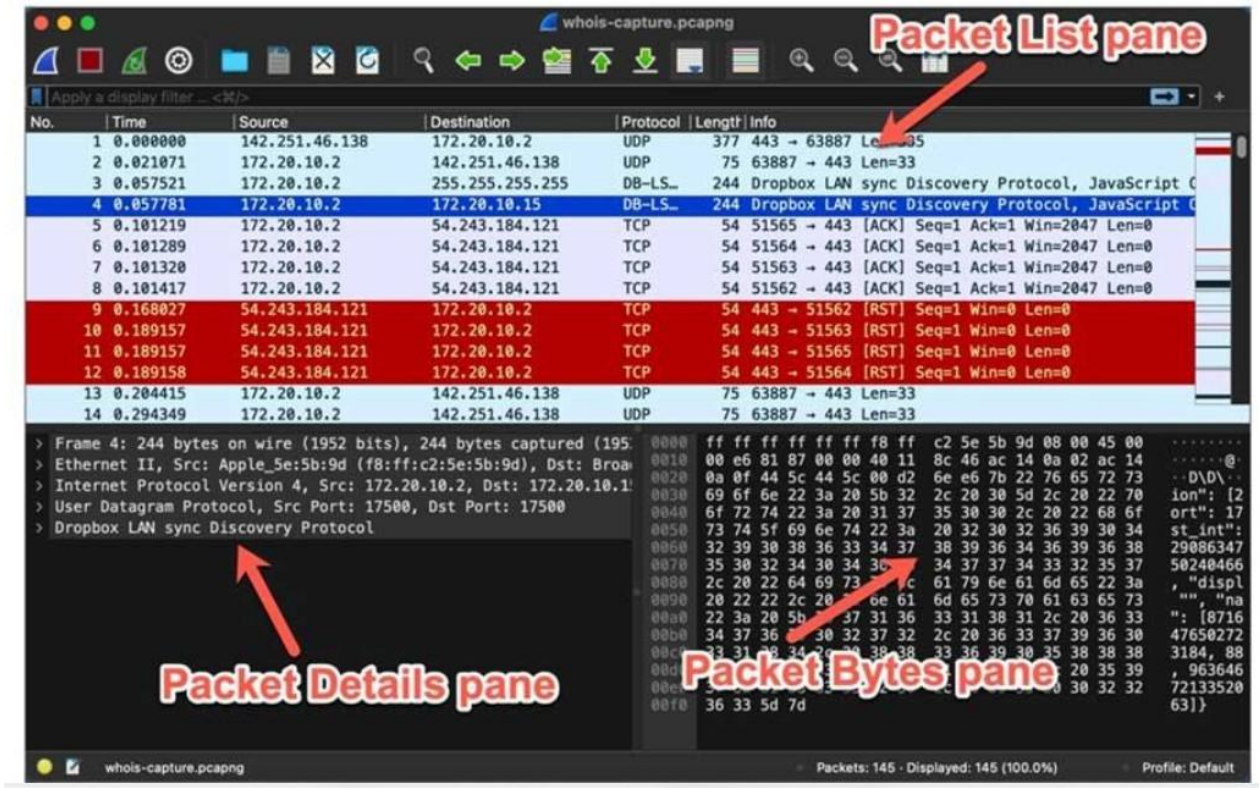


Step 3: Generate network traffic

While Wireshark is capturing:

1. Open a new browser tab.
2. Visit a website such as https://www.google.com.
3. Open two or three different websites.
4. Wait for approximately 30 seconds.
5. Return to Wireshark.
6. Click the red Stop button.

You should now see captured packets.



Step 4: Apply display filters

In the Wireshark filter bar, enter each filter separately and press Enter.

Filter	**Purpose**
tcp	Displays TCP packets
udp	Displays UDP packets
icmp	Displays ICMP packets
tcp.len > 0	Displays TCP packets carrying data

First, enter: TCP

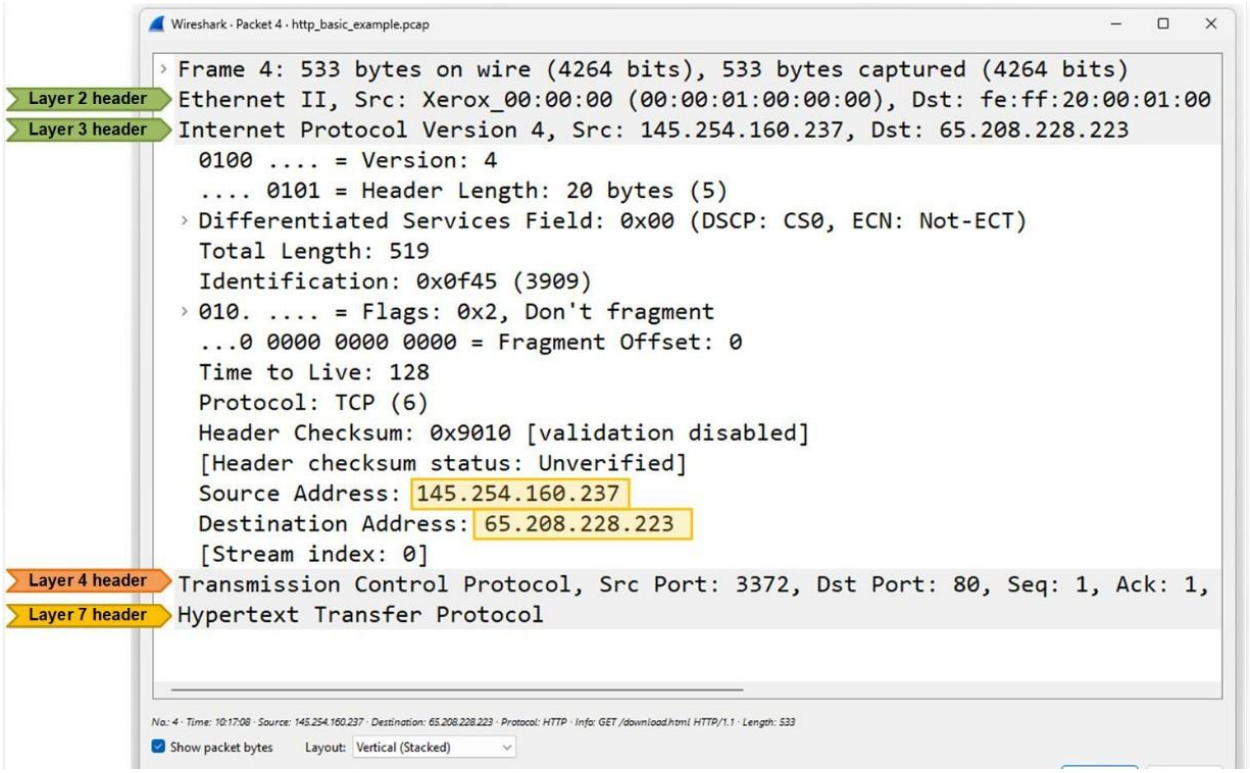
Observe the packets displayed.

Then enter: UDP

Observe the UDP packets.

Step 5: Inspect packet headers
Select any TCP packet from the packet list.

In the packet details pane, expand these sections:
1. Frame
2. Ethernet II
3. Internet Protocol Version 4
4. Transmission Control Protocol



Record the following values:

Parameter	Where to find it
Frame length	Frame
Ethernet Header	Ethernet II
Source MAC Address	Ethernet II
Destination MAC Address	Ethernet II
IPv4 Header Length	Internet Protocol Version 4
IPv4 Total Length	Internet Protocol Version 4
TCP Header Length	Transmission Control Protocol
TCP Segment Length	Transmission Control Protocol

Repeat the process for one UDP packet. Observe the UDP header and its length field.

Step 6: Calculate header overhead

For a basic Ethernet + IPv4 + TCP packet without TCP options:

- Ethernet header = 14 bytes
- IPv4 header = 20 bytes
- TCP header = 20 bytes
- Application data = 1500 bytes
- FCS = 4 bytes, if included in the frame representation

Total size including FCS:

$$14 + 20 + 20 + 1500 + 4 = 1558 bytes$$

Total overhead:

$$14 + 20 + 20 + 4 = 58bytes$$

Payload efficiency:

$$\frac{1500}{1558} \times 100 = 96.28\%$$

Note: Wireshark may not capture or display the Ethernet FCS. Use the actual captured packet fields when recording your observations.

Step 7:
Fill in your observation table

Record values from your own Wireshark capture.

Parameter	**TCP Packet**	**UDP Packet**
Frame Length	66	83
Ethernet Header Length	14	14
IP Header Length	20	20
Transport Header Length	32	8
IP Total Length	52	69
Transport Header Length	32	8
Payload Length	0	41

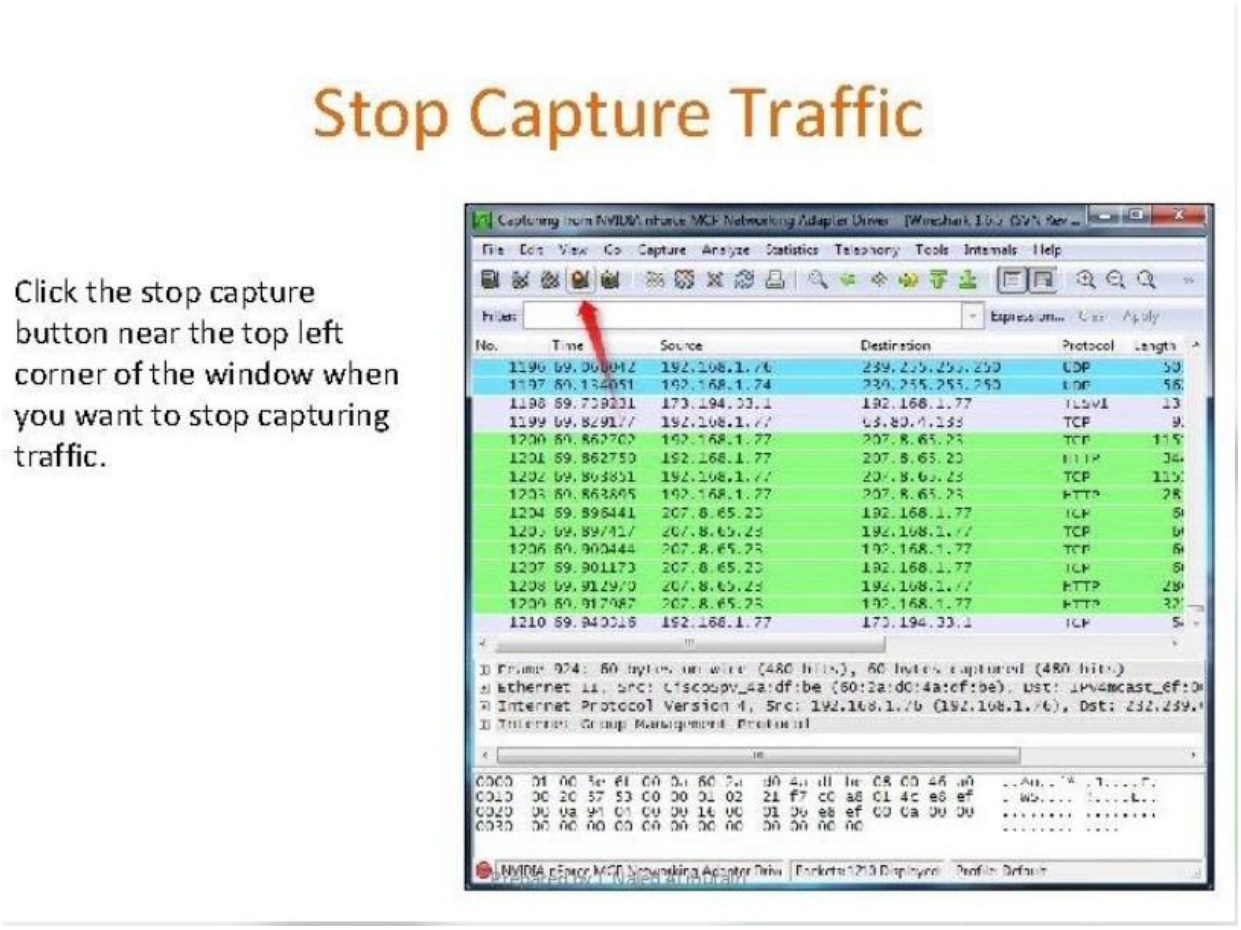
Steps to save a capture file

Step 1: Start capturing packets

1. Open Wireshark.
2. Select your Wi-Fi or Ethernet interface.
3. Double-click the interface to start capturing packets.

Step 2: Stop the capture

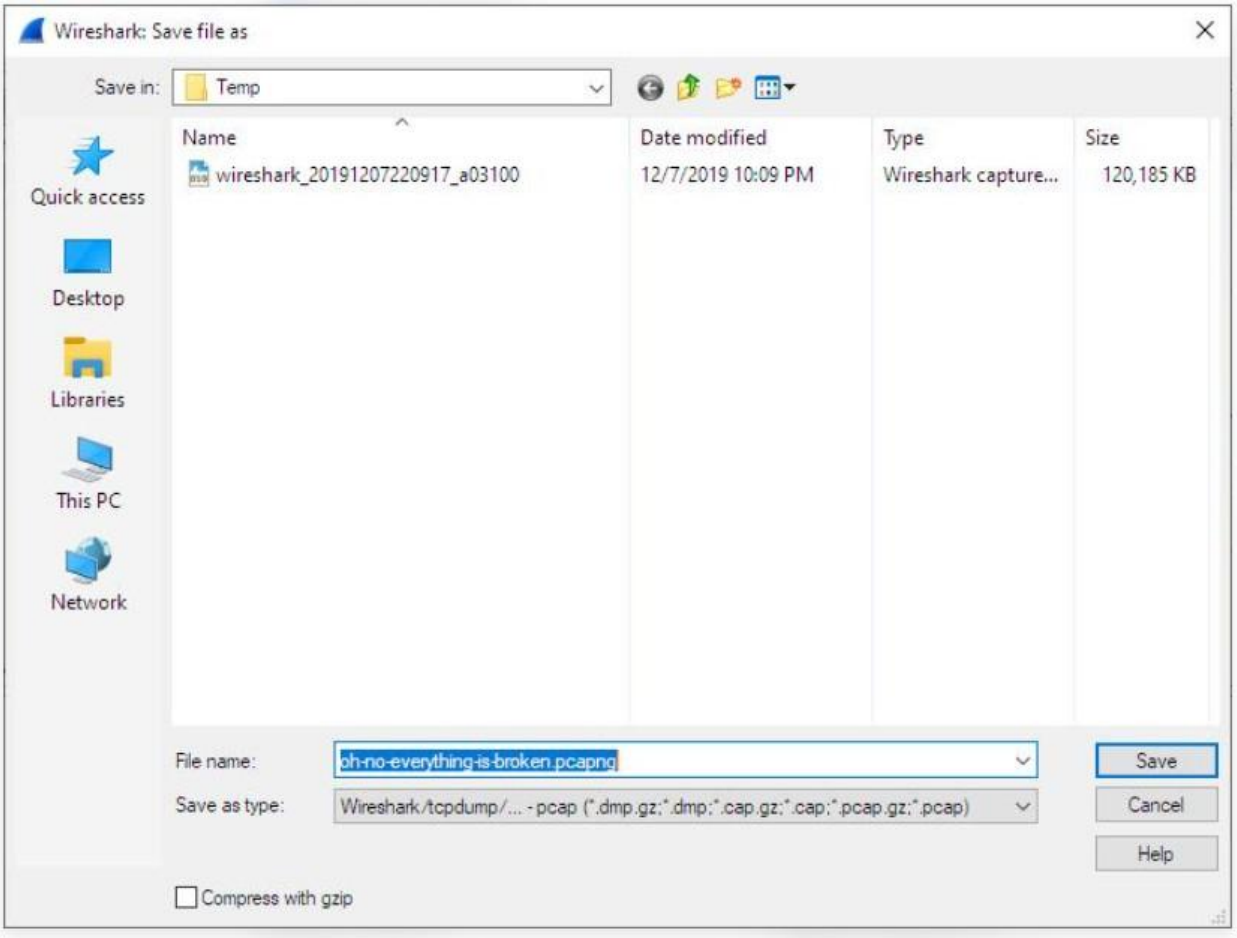
Once you have generated sufficient network traffic:
1. Click the red Stop button (top-left corner).
2. Your captured packets will remain visible on the screen.



Step 3: Save the capture

1. Click File in the menu.
2. Select Save As.
3. Choose the folder where you want to save the file.
4. Enter a filename.
5. Keep the file format as pcapng.
6. Click Save.

For example: sender_capture.pcapng



How to open a saved capture file

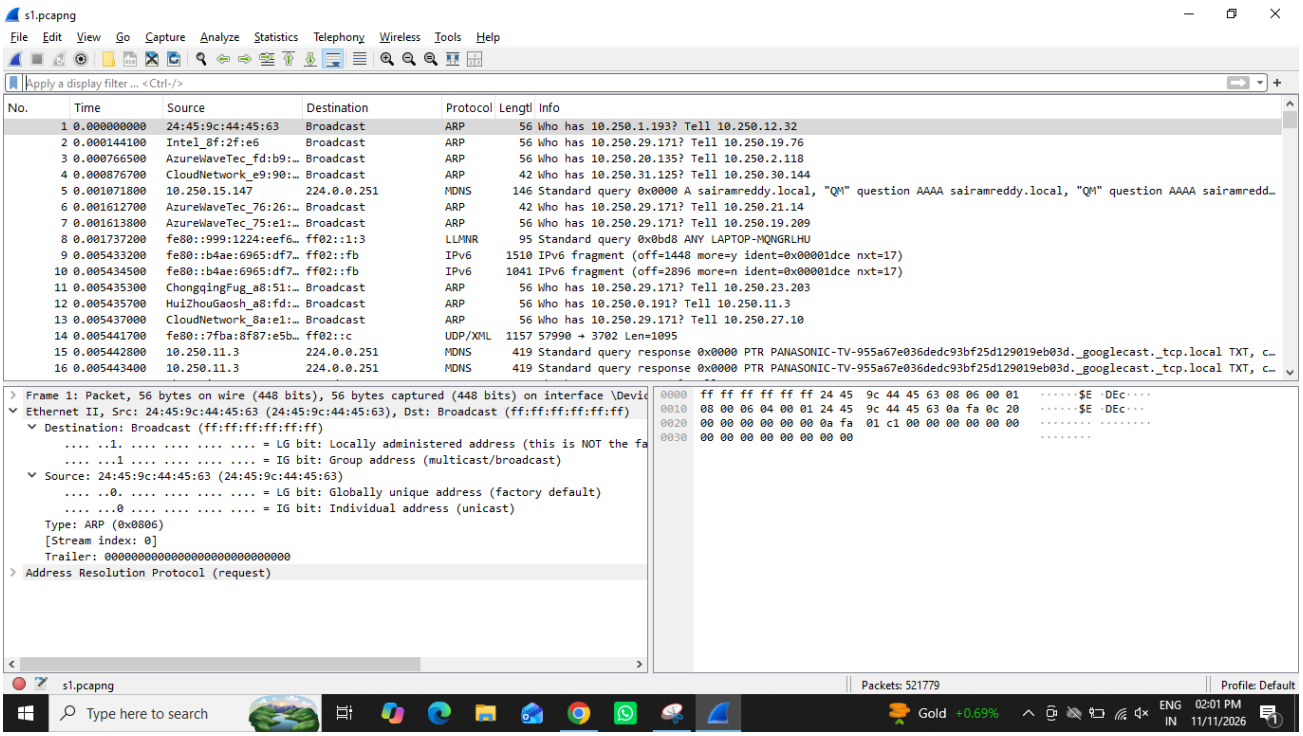
1. Open Wireshark.

2. Click File → Open.

3. Select your .pcapng or .pcap file.

4. Click Open.

5. Enter icmp in the display filter bar to view ICMP packets.



Answer these questions

1. What is encapsulation?

Ans. Encapsulation is the process where each layer of the OSI or TCP/IP model adds its own header (and sometimes trailer) to the data received from the layer above it before passing it down to the next layer. This prepares the data for transmission over the network.

2. What is the purpose of the Ethernet header?

Ans. It provides local Data Link layer addressing using Source and Destination **MAC addresses** and identifies the network layer protocol via the **EtherType** field.

3. What information is stored in the IP header?

Ans. It stores **Source and Destination IP addresses**, protocol version (IPv4/v6), TTL (Time to Live), Header/Total Length, and the upper-layer protocol identifier.

4. What is the difference between TCP and UDP headers?

- **TCP Header:** Larger ($20\text{--}60\text{ bytes}$), connection-oriented, and includes sequence numbers, ACKs, and flow control flags.
- **UDP Header:** Small (8 bytes fixed), connectionless, containing only ports, length, and checksum.

5. Why does each protocol add a header?

Ans. To attach specific control metadata needed by that layer for proper addressing, routing, sequencing, and data delivery across the network.

6. What is packet overhead?

Ans. Packet overhead is the extra protocol data (Ethernet, IP, and TCP/UDP headers) added to deliver the actual payload across the network.

7. How does header overhead affect network efficiency?

Ans. Higher overhead reduces data transfer efficiency because more bandwidth is consumed by headers rather than actual user payload. Smaller payloads result in lower efficiency, while larger payloads maximize efficiency.